

QIP 2025

PC report

Ángela Capel University of Cambridge, UK

Robert König Technical University of Munich

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The program committee

91 PC members

Scott	Aaronson	University of Texas
Gorjan	Alagic	University of Maryland
Alvaro	Alhambra	Spanish National Research Council
Anurag	Anshu	Harvard
Simon	Apers	CNRS, Université Paris Cité, IRIF
Srinivasan	Arunachalam	IBM Quantum
Ryan	Babbush	Google
Michael	Ben-Or	Hebrew University of Jerusalem
Michael	Beverland	IBM
Andreas	Bluhm	Université Grenoble Alpes
Sergey	Bravyi	IBM
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Ángela	Capel	University of Cambridge
Matthias C.	Caro	University of Warwick & Freie Universität Berlin
Ulysse	Chabaud	INRIA
Hao-Chung	Cheng	National Taiwan University
Giulio	Chiribella	QICI University of Hong Kong
Anirban	Chowdhury	IBM
Nicolas	Delfosse	IonQ
Eleni	Diamanti	CNRS & Sorbonne University
Frederic	Dupuis	Université de Montréal

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Uma	Girish	Columbia University
Jeongwan	Haah	Microsoft / Stanford
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Peter	Høyer	University of Calgary
Min-Hsiu	Hsieh	Foxconn
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Liang	Jiang	University of Chicago
Stephen	Jordan	Google
Elham	Kashefi	University of Edinburgh & CNRS - Sorbonne Université
Michael	Kastoryano	University of Copenhagen & AWS
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Robert	Koenig	Technical University of Munich
Tamara	Kohler	Stanford
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François	Le Gall	Nagoya University
Soojoon	Lee	Kyung Hee University

Anthony Lin	Leverrier Lin	INRIA UC Berkeley
Fermi	Ma	UC Berkeley
Laura	Mančinska	University of Copenhagen
Damian	Markham	Sorbonne Université
Dmitri	Maslov	Google
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Johannes Jakob	Meyer	Freie Universität Berlin
Tomoyuki	Morimae	Yukawa Institute for Theoretical Physics
Hamoon	Mousavi	UC Berkeley
Markus	Müller	Institute for Quantum Optics and Quantum Information, Vienna
Anand	Natarajan	MIT
Hui Khoon	Ng	National University of Singapore
Yingkai	Ouyang	University of Sheffield
Subhasree	Patro	Eindhoven University of Technology & CWI
Saleh	Rahimi-Keshari	Institute for Research in Fundamental Sciences
Daniel	Ranard	Caltech
Bartosz	Regula	RIKEN
Joseph M. Marc-Olivier	Renes Renou	ETH Zurich INRIA
Robert	Salzmann	École normale supérieure de Lyon
Changpeng	Shao	Academy of Mathematics and Systems Science, Chinese Academy of Sciences
Mehdi	Soleimanifar	Caltech

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David	Sutter	IBM
Mario	Szegedy	Rutgers University
Ryuji	Takagi	University of Tokyo
Eugene	Tang	Northeastern University
Nathanan	Tantivasadakarn	Caltech
Marco	Tomamichel	National University of Singapore
Rahul	Trivedi	Max-Planck-Institut für Quantenoptik
Peter	Vrana	Budapest University of Technology and Economics
Yuxin	Wang	University of Maryland
Xin	Wang	Hong Kong University of Science and Technology - Guangzhou
Albert H.	Werner	University of Copenhagen
Dominik	Wild	Max-Planck-Institut für Quantenoptik
Ramona	Wolf	Universität Siegen
Michael	Wolf	Technical University of Munich
Ronald	de Wolf	QuSoft, CWI & University of Amsterdam
John	Wright	UC Berkeley
Hayata	Yamasaki	University of Tokyo
Penghui	Yao	Nanjing University
Bei	Zeng	University of Texas at Dallas
Shengyu	Zhang	Tencent
Sisi	Zhou	Perimeter Institute for Theoretical Physics
Zoltan	Zimboras	Wigner Research Center for Physics

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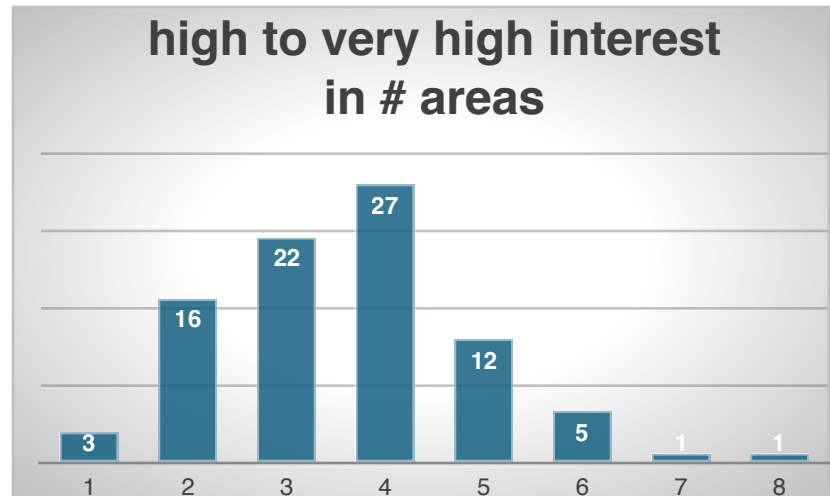
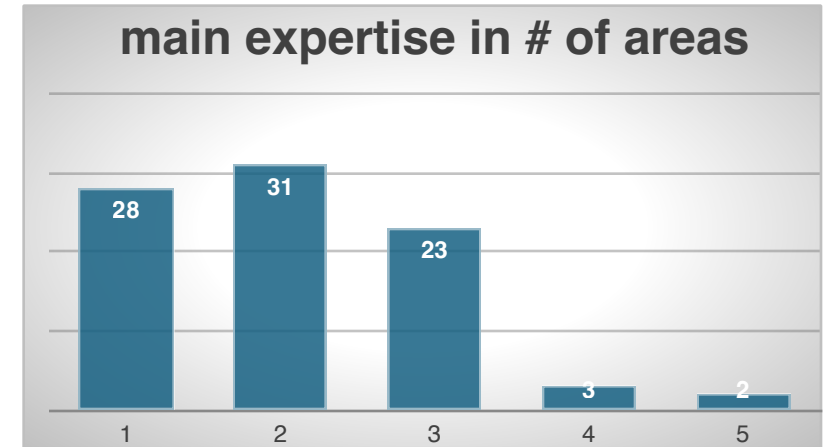
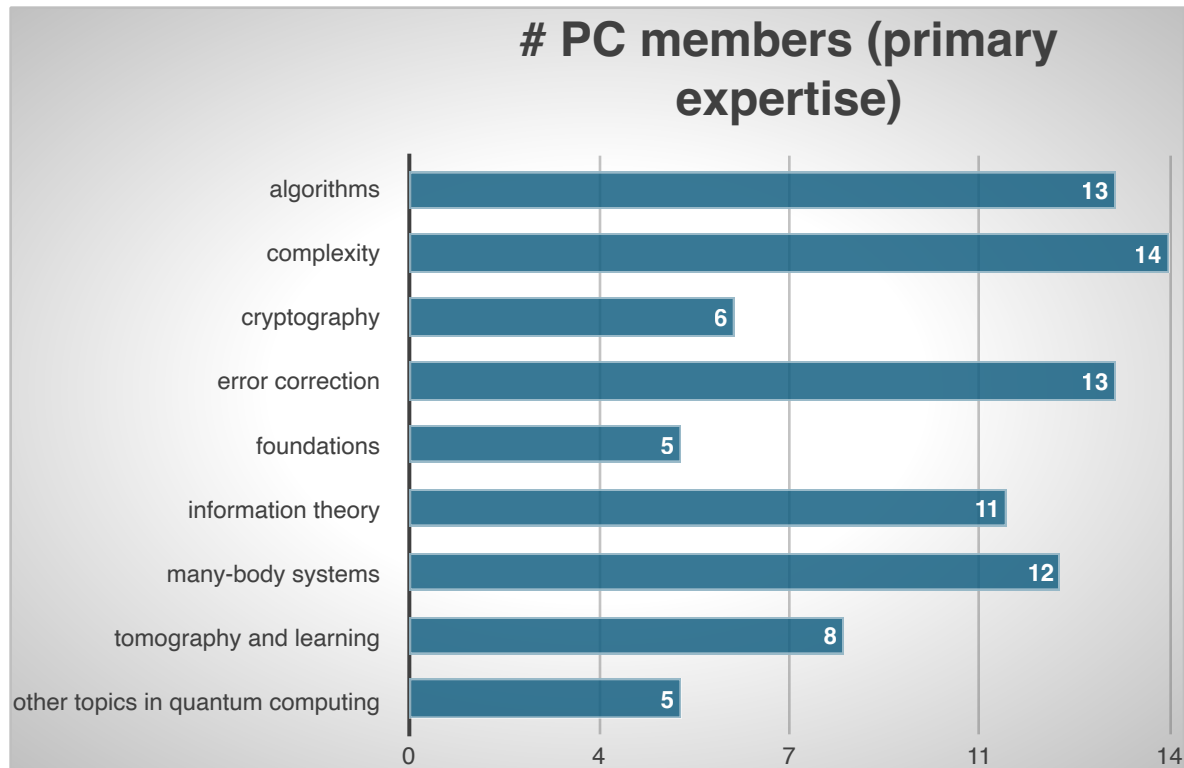
Lukasz	Fidkowski	University of Washington
Sevag	Gharibian	Universität Paderborn
Uma	Girish	Columbia University
Jeongwan	Haah	Microsoft / Stanford
Jonas	Haferkamp	Harvard
Peter	Høyer	University of Calgary
Min-Hsiu	Hsieh	Foxconn
Felix	Huber	Université de Bordeaux
Zhengfeng	Ji	Tsinghua University
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+ 886 subreviewers

PC: Expertise and interest



Submissions and acceptance rate

778 **submissions** to the talk track

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163 accepted submissions (151 talks due to merges)

acceptance rate: 20.1%

3 long plenary talks

3 long plenary talks invited by the Steering Committee

7 short plenary talks

138 talks in parallel sessions

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198 submissions to the poster-only track



197 accepted submissions

Submissions and acceptance rate

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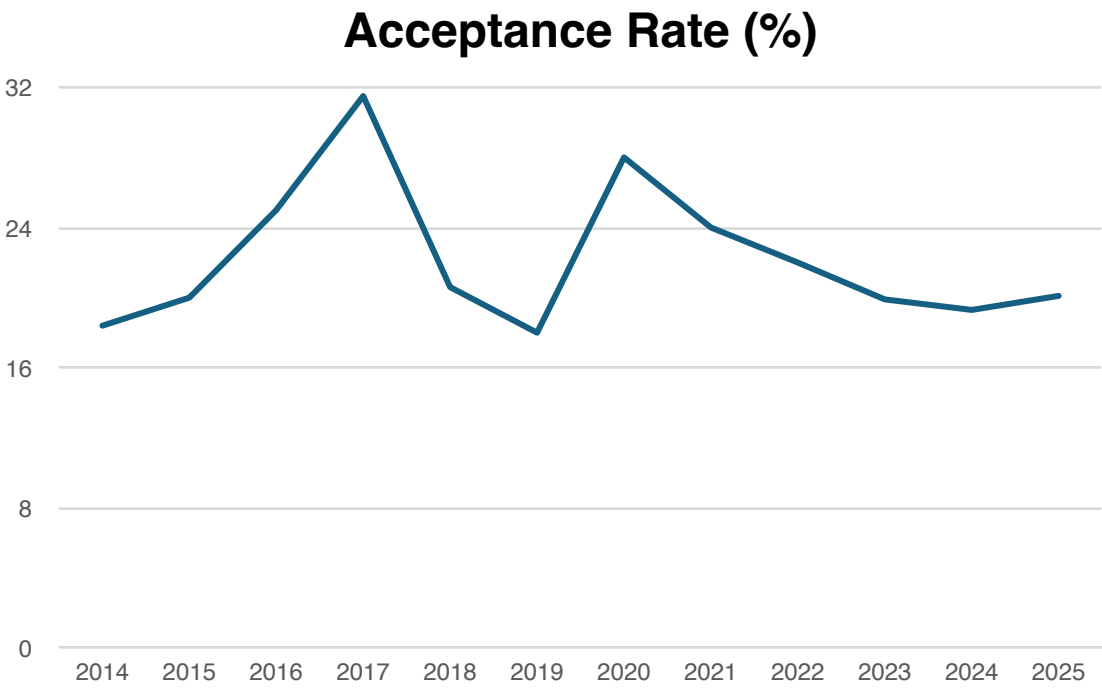
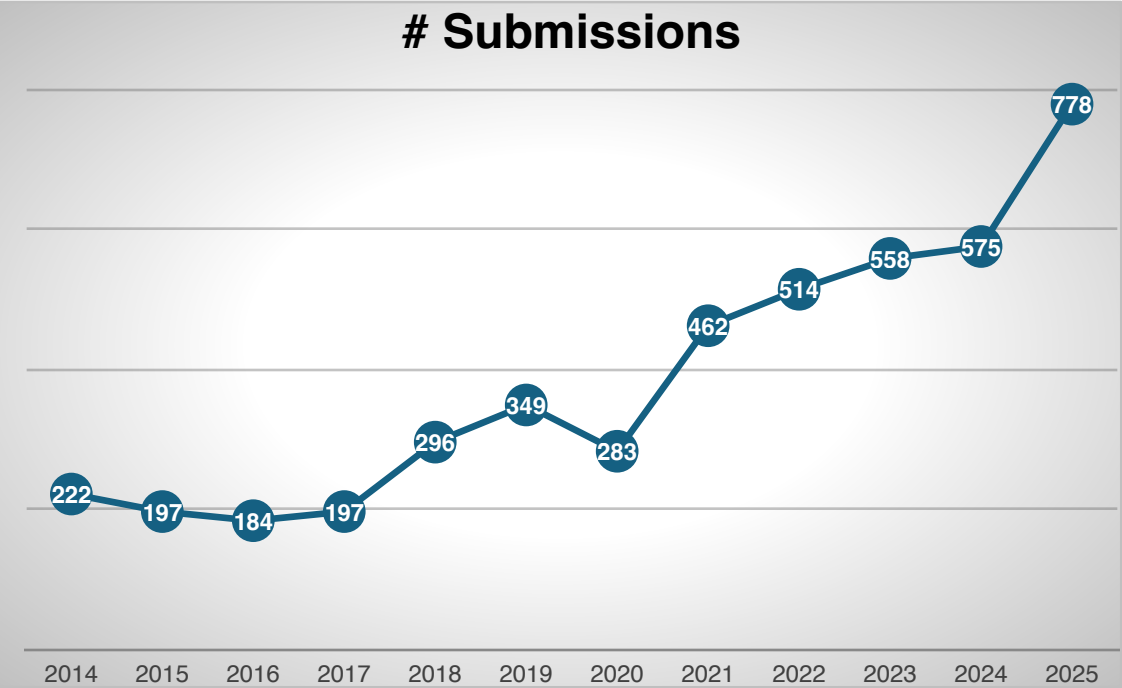
197 accepted submissions

812 Accepted posters in total



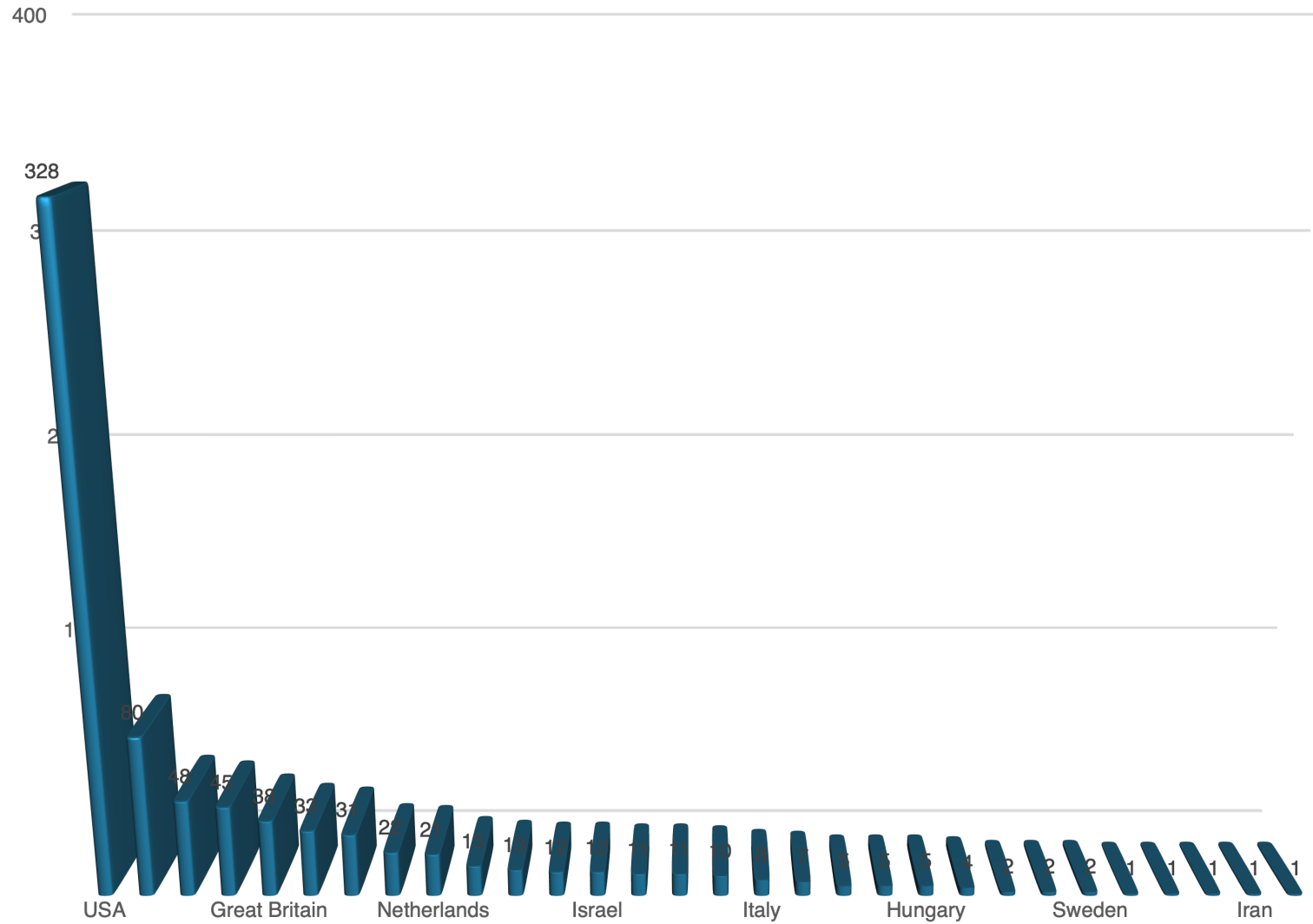
357 (only 44%) willing to be presented

QIP submissions and acceptance rates over the years

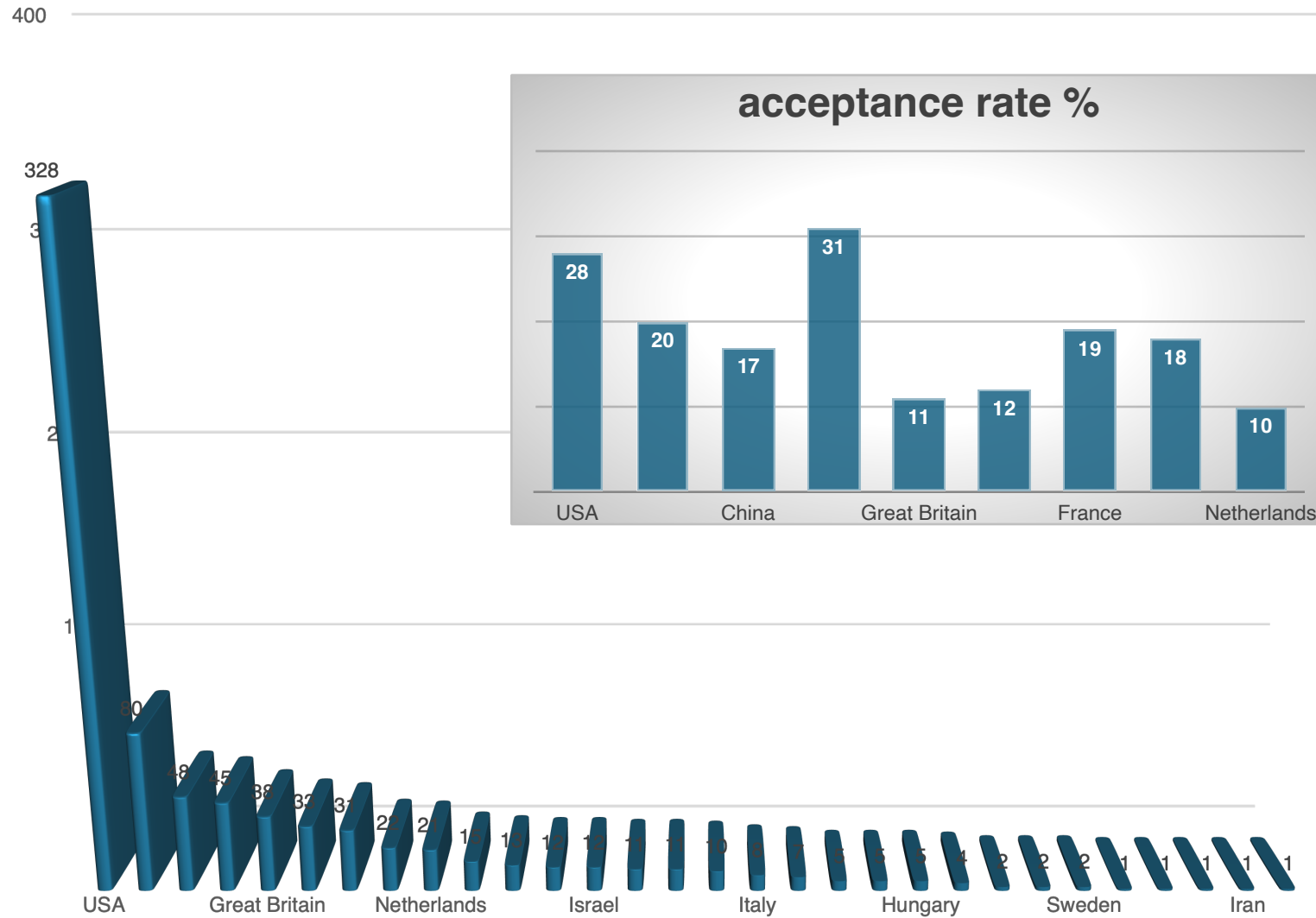


778	20.1%
# submissions	acceptance rate

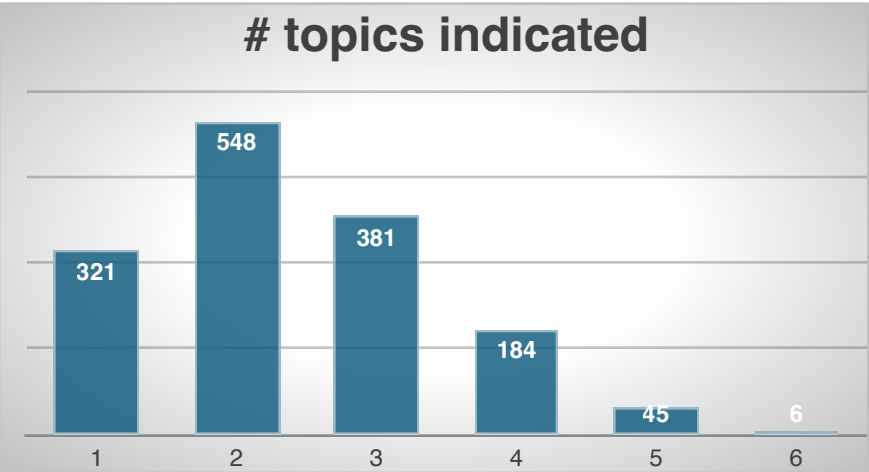
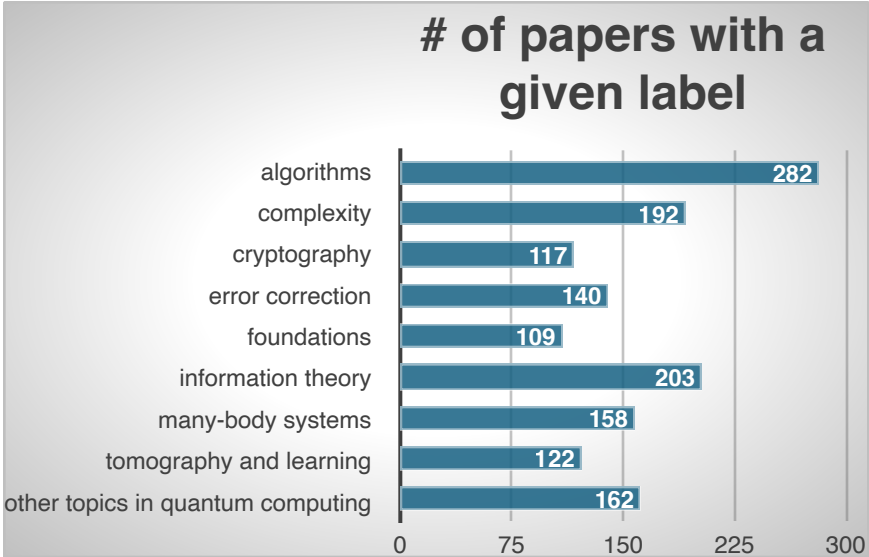
Submissions by country



Submissions by country

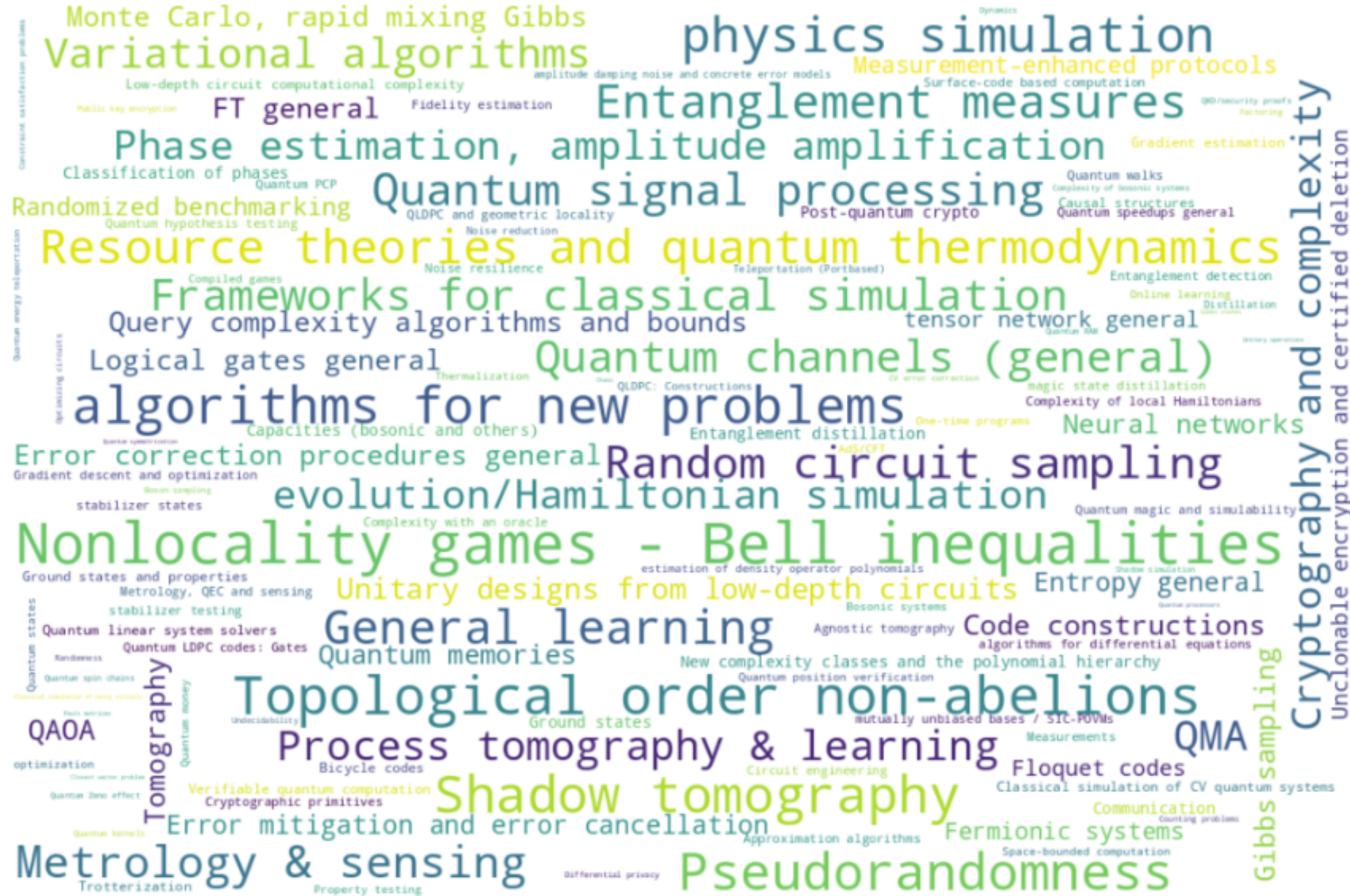


Topic categories



	algorithms	complexity	cryptography	error correction	foundations	information theory	many-body systems	tomography and learning	other topics in quantum computing
algorithms	0	100	13	23	18	47	63	53	64
complexity	100	0	35	12	23	30	42	29	29
cryptography	13	35	0	4	14	23	4	10	12
error correction	23	12	4	0	9	26	18	9	25
foundations	18	23	14	9	0	50	16	10	33
information theory	47	30	23	26	50	0	35	26	48
many-body systems	63	42	4	18	16	35	0	22	36

PC work



identified
145 initial clusters

use the Physics and Astronomy Classification Scheme (PACS) or similar at submission time?

PC work

Review phase

~ 4 weeks

Target

3 reviews per paper (99% reached)

Workload

27-28 papers per PC member

Longer review phase?

- *more confidence in correctness of results*
- *would require starting the entire process earlier.*

PC work

Review phase

~ 4 weeks

Target

3 reviews per paper (99% reached)

Workload

27-28 papers per PC member

Discussion phase:

< 4 weeks.

papers discussed

314 (40%)

average # of comments/paper

4.5

Longer review phase?

- *more confidence in correctness of results*
- *would require starting the entire process earlier.*

Difficult choices had to be made.

A lot of good work could not be accepted.

Public availability of technical versions

“Technical Manuscript: [...]”

Submissions whose technical manuscript is publicly available will be preferred. Public availability of the technical version a month after the submission deadline will factor in the Program Committee's choice of accepted talks.”

QIP 2025 call for papers

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QIP 2025 call for papers

457/778 (59%)

of submissions have
online technical versions



Oct 4

Submission
deadline

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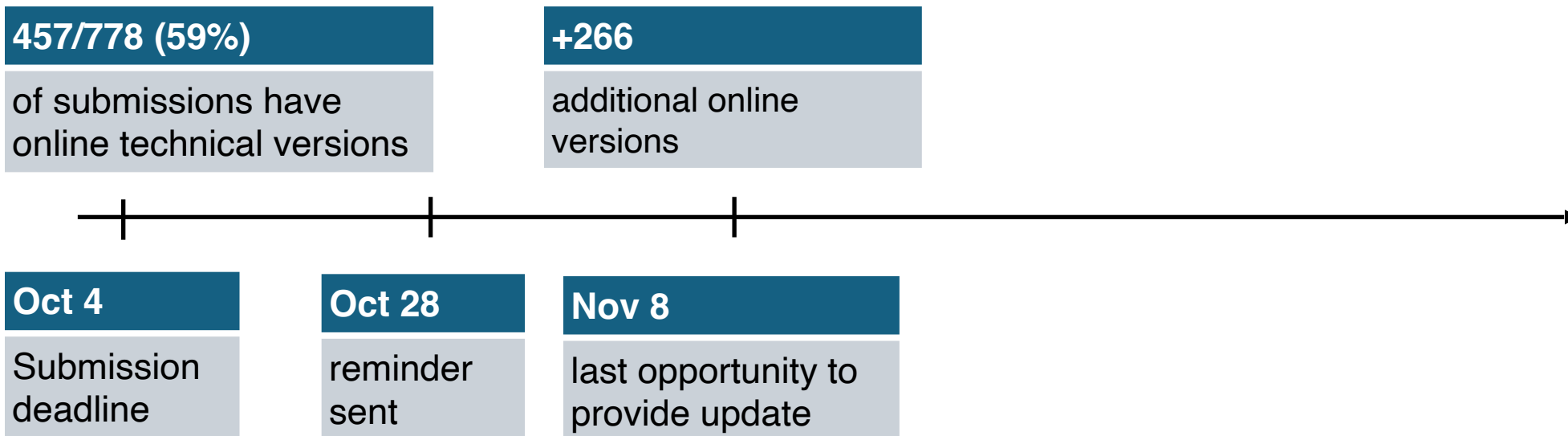


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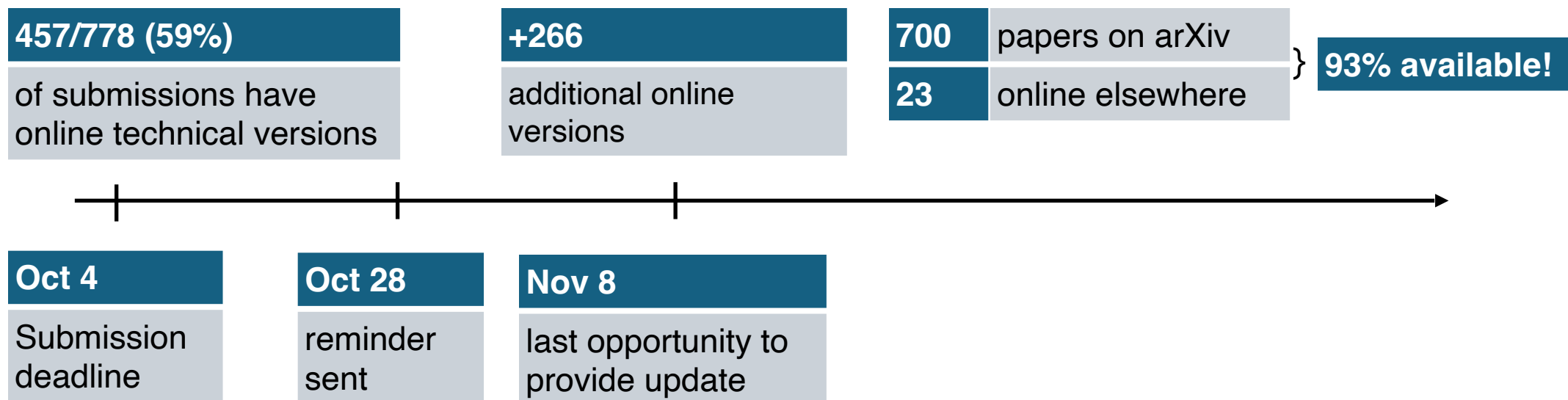


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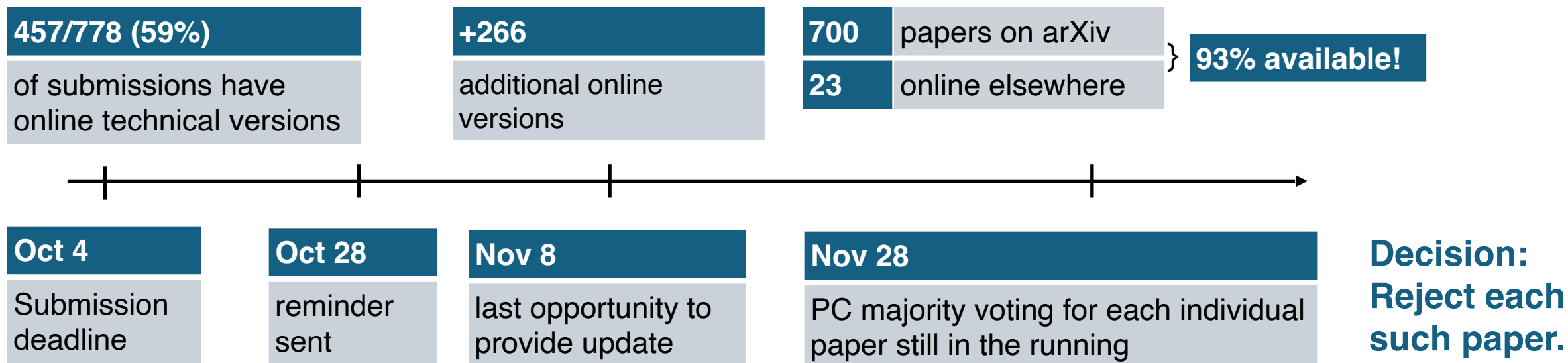


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QIP 2025 call for papers



Issues and a proposal for a revised call

- Fuzzy wording in call creates confusion and misunderstandings
-
- Increased complexity of submission handling for PC
-
- Questionable (incomplete or differing) arXiv-postings
-
- Postings on personal/other websites which are not easily locatable

Proposed strengthened wording:

“A technical version

- **is mandatory and**
- **can only be provided in the form of an arXiv-link.”**

QIP 2025 best student paper award

“A submission is eligible for the Best Student Paper prize if and only if the main author(s) is/are a student(s) at the time of the submission and will present the work at QIP, and further a significant portion of the work (at least 60%) has been done by said student(s), including contributing the majority of the key ideas.”

QIP 2025 call for papers



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317 (41%) papers eligible at submission



61 (37%) papers eligible in final program

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.....and the winner is:

QIP 2025 call for papers



QIP 2025 best student paper award

plenary talk tomorrow
(Thursday)

Constant-Overhead Magic State Distillation

Adam Wills (MIT; Hon Hai (Foxconn) Research Institute)
Min-Hsiu Hsieh (Hon Hai (Foxconn) Research Institute) Hayata Yamasaki (University of Tokyo)

Asymptotically Good Quantum Codes with Transversal Non-Clifford Gates

Louis Golowich (University of California at Berkeley)]
Venkatesan Guruswami (University of California at Berkeley)

Good binary quantum codes with transversal CCZ gate

Quynh T Nguyen (Harvard University)

Quantum LDPC Codes with Transversal Non-Clifford Gates via Products of Algebraic Codes

Louis Golowich (University of California at Berkeley)
Ting-Chun Lin (University of California at San Diego)



QIP 2025: gender distribution (male/female, estimated)

	Total # of female authors	Total # of authors	percentage of female authors
all submissions	275	1858	15%
accepted submissions	50	428	12%

(unique authors only,
i.e., excluding multiple
submissions)

Data analyzed and slide added after business meeting on request